ECON 770 Recitation 3

Aug 28, 2020

University of North Carolina at Chapel Hill

Reminder

- Due of Assignment #2: Sep 8, 2020 12:00 PM

Contents

- Homework Questions
- Continuous Random Variable
- Leibniz Integral Rule
- Random Variable Transformation

Part 1 - Question 2

Show that the set of irrational numbers $\mathbb{Q}^c$ are uncountable.

Proof.

SNOT, i.e. $\mathbb{Q}^c$ is countable then $\mathbb{Q}^c$ on $(0,1)$ is countable. Since $\mathbb{Q}$ on $(0,1)$ is countable, then $(0,1)$ as a union is countable. Therefore, we can write all numbers with infinitely many digits $(0,1) = \{x_1, x_2, \dots\}$ where

$$\begin{aligned} x_1 &= 0.a_{11}a_{12}a_{13}\dots \\ x_2 &= 0.a_{21}a_{22}a_{23}\dots \\ x_3 &= 0.a_{31}a_{32}a_{33}\dots \end{aligned}$$

where $a_{ij} \in \{0,1,2,3,\dots,9\}$ and there are infinitely many nonzero a_{ij} in x_i. Define $x_0 = 0.b_1b_2b_3\dots$ s.t.

$$b_i = \begin{cases} 1 & \text{if } a_{ii} \neq 1 \\ 2 & \text{if } a_{ii} = 1 \end{cases}$$

Then $x_0 \notin (0,1)$, contradiction. $\square$

Part 1 - Question 3

How to find a σ-field from a **set** A?

- $\sigma(A) \not\equiv \sigma$-field in Ω
- $\sigma(A) \not\equiv \mathcal{P}(\Omega)$ (and obviously $\sigma(A) \not\equiv \mathcal{P}(A)$)
- By definition, $\varnothing \in \sigma(A)$, so $\Omega \in \sigma(A)$
- By definition again, $A \in \sigma(A) \implies A^c \in \sigma(A)$
- Done with $\sigma(A) = \{\varnothing, A, A^c, \Omega\}$

Part II - Question 2

Prove **axiomatically** $P(\cap_{i=1}^{\infty} A_i) \leq \liminf_{i\to\infty} P(A_i)$

Proof.

Suppose that $\{A_i\}_{i=1}^{\infty}$ is a sequence of events, then by definition,

$$\liminf_{i\to\infty} P(A_i) \equiv \sup_{n\geq 1} \inf_{m\geq n} P(A_m) \geq \inf_{m\geq 1} P(A_m)$$

Since $\cap_{i=1}^{\infty} A_i \subset A_k \ \forall\, k \in \mathbb{N}$, then we know $P(\cap_{i=1}^{\infty} A_i) \leq P(A_k) \ \forall\, k \in \mathbb{N}$ (Show this for any finite k by constructing sets). Then, we claim that $P(\cap_{i=1}^{\infty} A_i) \leq$ [a] $\inf_{k\geq 1} P(A_k)$. □

[a] SNOT, then $\exists\ \varepsilon > 0$ s.t. $P(\cap_{i=1}^{\infty} A_i) \geq \inf_{k\geq 1} P(A_k) + \varepsilon \equiv p$, and show $\exists\ k' s.t. P(A_{k'}) < p$, contradiction*

Continuous Random Variable

Defn.

Random Variable $X(\omega)$ is continuous if support S is uncountable. Continuous CDF is defined as $F(x) = P(X \leq x) = \int_{-\infty}^{x} f(t)dt$ where $f(x)$ is the probability density function (PDF). As a result, $f(x)$ satisfies

- $f(x) \geq 0 \ \forall x$
- $\int_{-\infty}^{\infty} f(x)dx = 1$ or $\int_{x \in S} f(x)dx = 1$

Prop.

- 0-Mass:
  $P(X = a) = \lim_{\varepsilon \to 0} P(a \leq X \leq a + \varepsilon) = \lim_{\varepsilon \to 0} \int_{a}^{a+\varepsilon} f(x)dx = 0$
- $P(a \leq X \leq b) = F(b) - F(a) = \int_{a}^{b} f(x)dx$

Leibniz Integral Rule

Theorem

Given integral $\int_{a(x)}^{b(x)} f(x,t)dt$, *where* $a(x), b(x) \in (-\infty, \infty)$, *then the derivative of this integral is expressible as*

$$\frac{d}{dx}\left(\int_{a(x)}^{b(x)} f(x,t)dt\right) = f(x,b(x)) \cdot \frac{d}{dx}b(x) - f(x,a(x)) \cdot \frac{d}{dx}a(x) + \int_{a(x)}^{b(x)} \frac{\partial}{\partial x} f(x,t)dt$$

If both $a(x)$ *and* $b(x)$ *are constant, then*

$$\frac{d}{dx}\left(\int_{a}^{b} f(x,t)dt\right) = \int_{a}^{b} \frac{\partial}{\partial x} f(x,t)dt$$

Random Variable Transformation

Let $X \sim P(\cdot)$ on support $S_x \subset \mathbb{R}$, define $Y = g(X)$ for some "measurable" $g : S_x \to S_y \subset \mathbb{R}$

- Discrete R.V.
  - Strictly monotonic g (thus 1-1 mapping, e.g. $g(x) = 2x + 5$)

$$P(Y = g(x)) = P(X = x)$$

  - Non-Monotonic g (thus we may have x_1, x_2 s.t. $g(x_1) = g(x_2)$

$$P(Y = g(x)) = P(X \in \{x' \in S_x | \, g(x') = g(x)\})$$

- Continuous R.V.
  - Strictly monotonic g (thus $\exists \, g^{-1} : S_y \to S_x$ s.t. $g^{-1}g = 1$)

$$P(Y \leq y) = P(g(X) \leq y) = \begin{cases} P(X \leq g^{-1}(y)) & \text{if } g \uparrow \\ P(X > g^{-1}(y)) & \text{if } g \downarrow \end{cases}$$

$$f_Y(y) = \frac{\partial}{\partial y} F_Y(y) = f_X(g^{-1}(y)) \left| \frac{\partial g^{-1}(y)}{\partial y} \right|$$

Random Variable Transformation (Contd)

Formally, we have a theorem for monotonic transformation of continuous R.V.

Theorem

Let X be a continuous R.V. with density f_X, let $Y = g(X)$. If

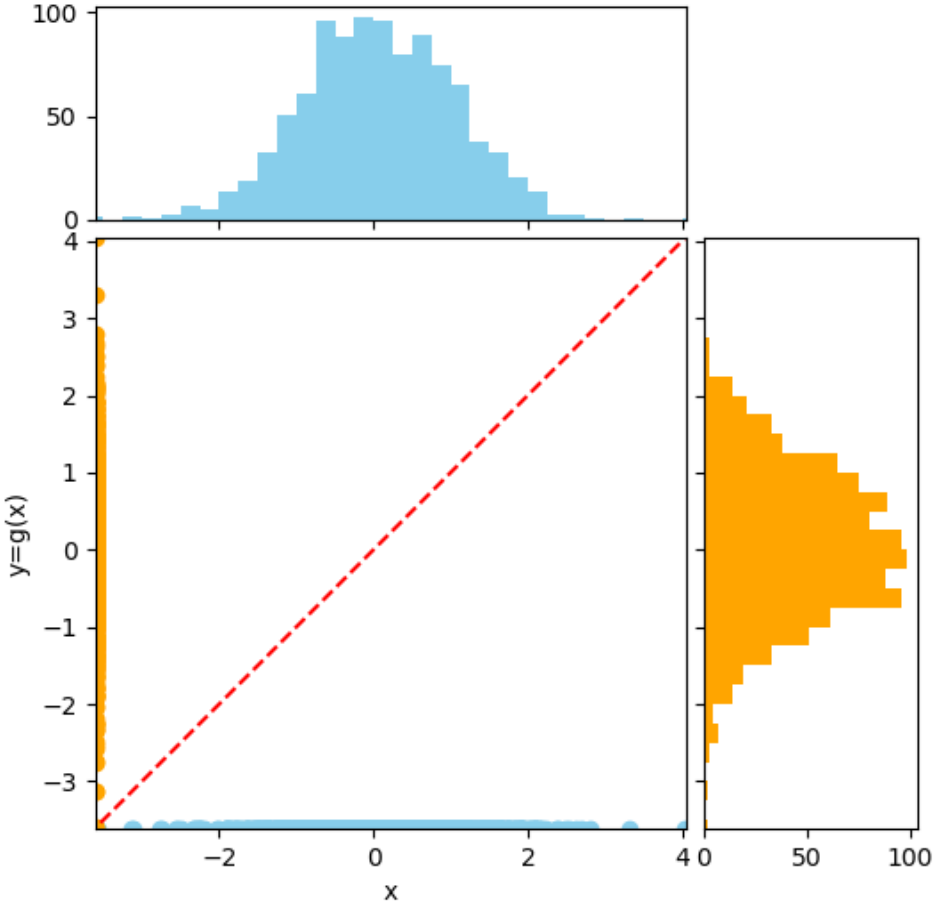
1. *$g(\cdot)$ is monotone.*
2. *$f_X(x)$ is continuous on $S_X \subset \mathbb{R}$*
3. *$g^{-1}(y)$ is continuously differentiable on $S_Y \subset \mathbb{R}$*

then, PDF of Y is given by

$$f_Y(y) = \begin{cases} f_X\left(g^{-1}(y)\right)\left|\frac{d}{dy}g^{-1}(y)\right| & \text{if } y \in S_Y \\ 0 & \text{if } y \notin S_Y \end{cases}$$

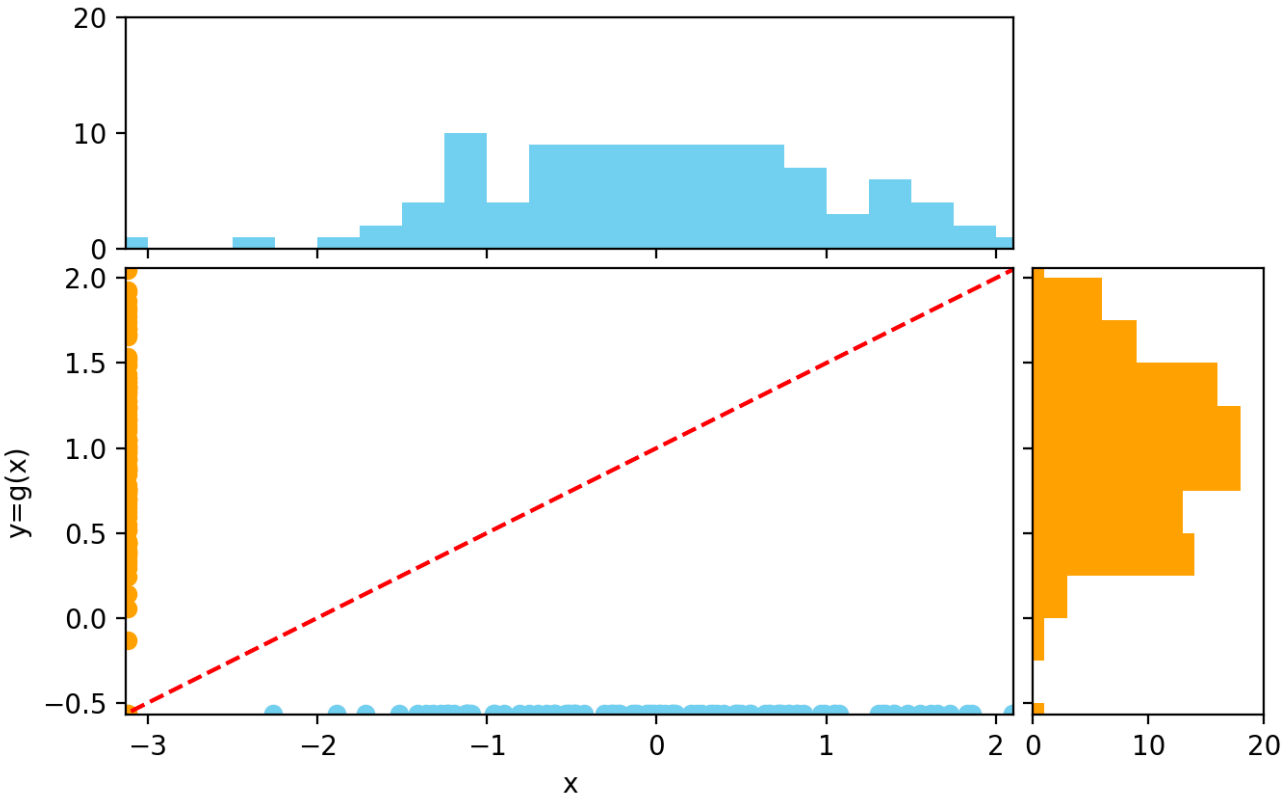
Random Variable Transformation (Illustration)

Figure: Transformation $y = g(x) = x$



Random Variable Transformation (Illustration)

Figure: Transformation $y = g(x) = 0.5x + 1$



Random Variable Transformation (Contd)

What if g is not strictly monotonic?[1] Split it.

Theorem

Let X be a continuous R.V. with continuous density f_X. Let $A_0, A_1, \ldots, A_k$ be a partition of S_X s.t. $P(X \in A_0) = 0$ and $P(X \in A_i) > 0$ $(i = 1, \ldots, k)$. Define $g_i(x)$ on A_i for $i = 1, \ldots, k$, respectively. If

1. $g(x) = g_i(x)$ *for* $x \in A_i$
2. $g_i(x)$ *is monotonic on* A_i
3. $g_i^{-1}(y)$ *is continuously differentiable on* $S_{Yi} = g_i(A_i)$

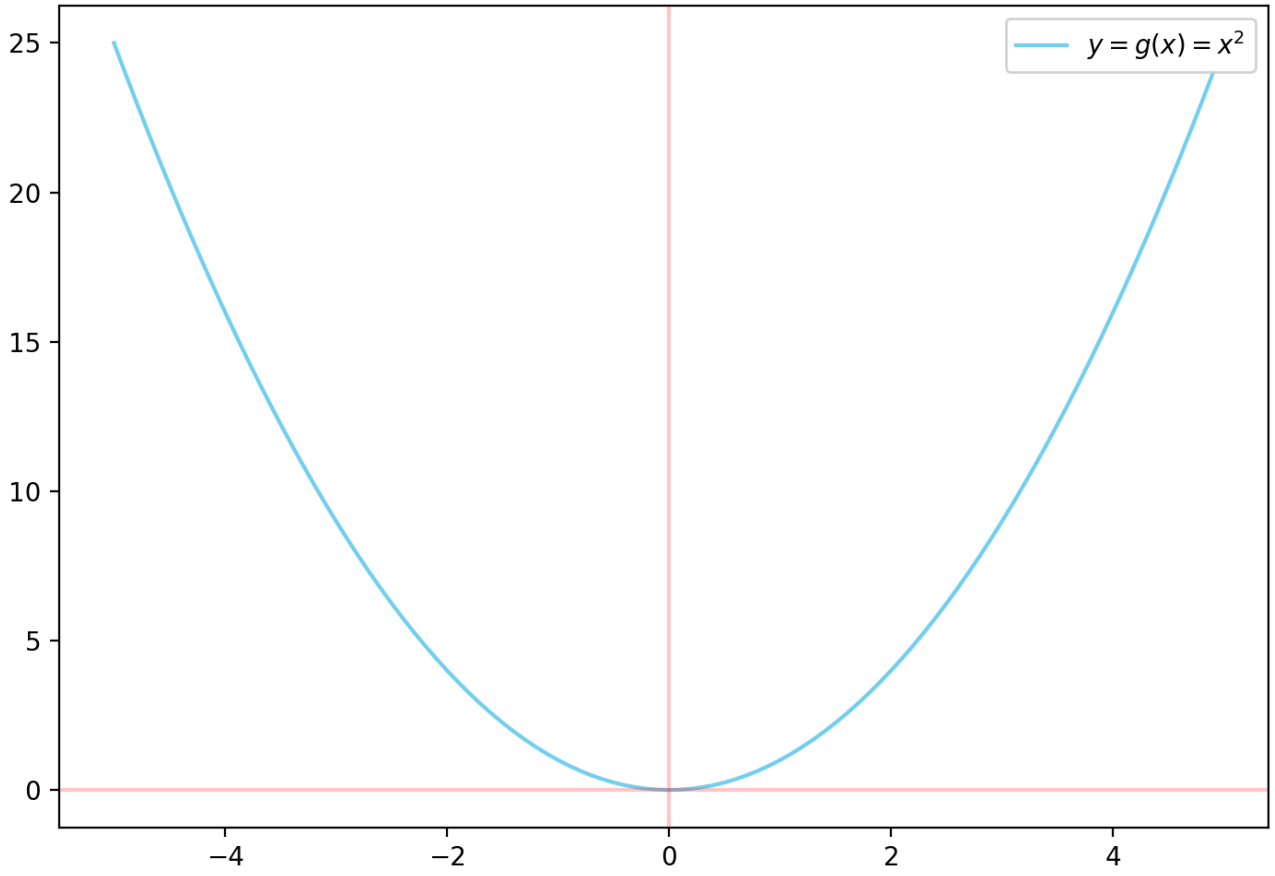
Then

$$f_Y(y) = \sum_{i=1}^{k} f_Y^{(i)}(y) = \sum_{i=1}^{k} \mathbb{1}_{\{y \in S_{Yi}\}} f_X\left(g_i^{-1}(y)\right) \left| \frac{d}{dy} g_i^{-1}(y) \right|$$

[1] You may do it by cases, see lecture notes

Random Variable Transformation (Illustration)

Figure: Transformation $y = g(x) = x^2$



Random Variable Transformation (Proof)*

Proof.

For any interval $I = (a, b] \subset \mathbb{R}$, if $I \cap S_Y = \varnothing$, then $P(a < Y \leq b) = 0$. If not, then,

$$\{Y \in I \cap S_Y\} = \left\{Y \in I \cap \left(\cup_{i=1}^{k} S_{Yi}\right)\right\} = \left\{Y \in \cup_{i=1}^{k} \left(I \cap S_{Yi}\right)\right\} \equiv \{X \in \cup_{i=1}^{n} A_i'\}$$

where $A_i' \subseteq A_i$ for all i and hence $A_i' \cap A_j' = \varnothing$ if $i \neq j$. Therefore,

$$P(a < Y \leq b) = P(Y \in I \cap S_Y) = \sum_{i=1}^{n} P(X \in \cup_{i=1}^{n} A_i') = \sum_{i=1}^{n} \int_{A_i'} f_X(x)dx$$

Since on each A_i', we have $y = g_i(x) \;\Leftrightarrow x = g_i^{-1}(y)$, then

$$P(a < Y \leq b) = \sum_{i=1}^{n} \int_{I \cap S_{Yi}} f_X\left(g_i^{-1}(y)\right) \left|\frac{d}{dy} g_i^{-1}(y)\right| dy$$